

TinyFLChain: Blockchain-Secured Decentralized Federated Learning for TinyML Systems

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Abstract—The rise of embedded AI has introduced critical challenges, including limited RAM, storage, and energy consumption. Emerging trends, such as TinyML, offer promising solutions to these constraints. In this work, we introduce TinyFLChain, a decentralized framework that combines TinyML, Federated Learning, and blockchain to enable privacy-preserving and secure collaborative learning on low-power IoT devices. By leveraging FL, computations are decentralized and data remains on local nodes, while blockchain ensures the integrity and traceability of model updates through smart contracts. Our framework utilizes blockchain smart contracts to secure the exchange of models updates, and implement a performance-based incentive mechanism. Clients who provide high-quality updates are rewarded at each round, while less active are penalized, encouraging sustained engagement. Using Raspberry Pi devices, we evaluated TinyFLChain on three benchmark datasets (MNIST, CIFAR-10, Digits). In centralised system, the accuracy reaches 100% on MNIST, 87% on CIFAR-10, and 97% on Digits. In decentralised system without blockchain, the performance is 99.25% on MNIST, 38.9% on CIFAR-10, and 10.72% on Digits. With TinyFLChain, accuracy significantly improves on CIFAR-10, reaching 62.09% for the best client, while remaining stable on MNIST. The results show that our system maintains high accuracy while improving security, transparency, and robustness against attacks.

Index Terms—Blockchain, Decentralized Federated Learning, TinyML, Internet of Things, Incentive Mechanism.

I. INTRODUCTION

The Internet of Things (IoT) is a rapidly evolving technology that is transforming the way devices interact with each other and the external world. In this context, researchers and professionals are constantly working to adapt and improve technologies to fully leverage these systems. Integrating machine learning (ML) and deep learning (DL) into IoT has helped address some of these challenges by providing solutions for data analysis directly on the devices themselves.

Integrating these technologies into IoT devices presents significant challenges, particularly due to hardware resource limitations. This is where TinyML proves valuable, enabling the execution of machine learning models on IoT devices while minimizing energy consumption and operating within constrained storage and computing capacities [1]. Moreover, federated learning and blockchain have emerged as key solutions to address IoT-related challenges, including security, privacy, and decentralized data management [2]. Federated

learning enhances data privacy while optimizing the learning process, whereas blockchain ensures secure transactions and transparency in data exchanges within IoT systems.

In this context, our work proposes TinyFLChain, an innovative architecture that combines Federated Learning, Blockchain, and TinyML to overcome the limitations of IoT devices while ensuring efficient model management and resource allocation. Our main contributions are as follows:

- Development of a secure Federated Learning architecture integrating Blockchain to ensure the integrity and traceability of trained models.
- We have implemented an incentive mechanism that dynamically identifies and rewards the best clients, encouraging quality participation.
- Optimization of communication between resource-constrained IoT devices through the use of lightweight communication protocols such as MQTT.
- Evaluation of TinyFLChain our approach on multiple datasets (MNIST, CIFAR-10, Digits) to evaluate the influence of dataset complexity on convergence and model performance.
- Identification of resource management challenges in a decentralized environment, highlighting hardware limitations and the need for resource optimization to ensure scalability.

The remainder of this paper is structured as follows. Section II presents the main concepts related to this paper, mainly blockchain, federated learning, IoT, TinyML, and MQTT. Section III provides a comprehensive overview of related work in decentralized federated learning in IoT. Section IV outlines the architecture of our proposed system. In section V, we outline details about the used datasets, and implementation specifics—along with the hardware configuration used by our system. Section VI presents experimental evaluations that demonstrate the system's effectiveness in improving security and resilience. Finally, Section VII concludes the work and outlines future directions.

II. BACKGROUND

This section provides a foundational overview of the core technologies discussed in this paper. We will highlight key technologies such as Federated Learning, Blockchain, IoT,

the MQTT protocol, and TinyML, emphasizing on tools like TensorFlow Lite. Together, these technologies facilitate the development of distributed, secure, and intelligent systems capable of processing sensitive data while fulfilling IoT environment's performance and security requirements.

Federated learning is a distributed way to train machine learning models that enable collaborative learning without requiring data transfer from client devices to global servers [3]. Instead, models are trained locally on edge devices using raw data, which improves data privacy. The final model is obtained by aggregating local updates.

Blockchain is a decentralized and immutable distributed ledger shared across a network of nodes [4]. While most commonly associated with cryptocurrencies, blockchain can be applied to any industry to ensure data integrity. Data is stored in blocks, linked chronologically to form a chain, with cryptographic techniques ensuring security [5]. Blockchain's consensus mechanism ensures the network validates transactions before being added to the ledger. This process involves recording a transaction as a data block, reaching a consensus on its validity, and then storing it securely, with any tampering detectable through altered cryptographic hashes. The updated ledger is then distributed across the network, ensuring transparency and trust.

The Internet of Things (IoT) is a network of interconnected devices that connects physical objects with digital entities through standardized electronics and wireless systems [6]. This enables seamless data exchange between the physical and virtual worlds, significantly improving areas like education, healthcare, security, and transportation. IoT also enhances enterprise productivity by providing a distributed network of smart devices, optimizing resources, and enabling cost savings.

MQTT (Message Queuing Telemetry Transport) is an open-source messaging protocol designed for efficient machine-to-machine communication, particularly in IoT systems with limited resources and bandwidth [7]. It uses a lightweight message structure with three Quality of Service (QoS) levels to balance transmission efficiency and message reliability, enabling clients to publish and receive messages through an MQTT broker.

TinyML combines machine learning with embedded systems like microcontrollers, enabling low-latency, low-power, and low-bandwidth inference at the edge [8]. The process begins by selecting an effective algorithm and gathering data, followed by model training on resource-rich devices. Afterward, the model is enhanced and ported to a microcontroller using TinyML interpreters, such as TensorFlow Micro, for inference on sensor data.

III. RELATED WORK

TinyML enables low-power devices to process data locally, which reduces latency, saves energy, and enhances privacy [9]. By using microcontrollers and optimized algorithms, IoT devices can make decisions independently without relying on cloud infrastructure. Federated TinyML goes greater by

allowing these devices to not only perform inference but also train models locally, preserving data privacy [10].

Llisterri et al. [11] explore training machine learning models on microcontrollers using federated learning, applied to a keyword detection task with Arduino Nano 33 BLE Sense boards. They analyze trade-offs between accuracy, resource use, and training time under various federated learning settings, concluding that more frequent rounds speed up loss reduction but increase bandwidth use and training duration.

Similarly, Ficco et al. [12] propose a method for locally training machine learning algorithms on resource-constrained IoT devices by combining federated learning with transfer learning. To communicate between devices and a central server, the MQTT protocol is used and C-optimized neural network libraries to reduce memory usage. The results show that the combination of FL and TL achieves an accuracy of 86.48% for classification tasks. Surpassing traditional TensorFlow Lite-based approaches.

Federated learning (FL) and blockchain are revolutionary approaches to improving the security of distributed artificial intelligence systems [13]. This combination is particularly relevant: TinyML, which implements machine learning with limited resources, and the Internet of Things (IoT), where decentralized and secure solutions are needed to manage large networks of connected devices.

The authors of this article [14] explore the use of Swarm Learning (SL), a decentralized federated learning system, combined with blockchain and TinyML. This approach allows low-resource devices to collaborate to drive a common model without a central server. The blockchain guarantees the security of parameter exchanges and protects devices against adversarial attacks. However, the article highlights a major challenge: the high computational cost of blockchain, which can slow down TinyML devices and limit their efficiency.

This article [15] introduces FLchain, a framework that integrates blockchain into federated learning to improve scalability and security of IoT networks. They identify several challenges related to communication latency, resource allocation and incentive management to encourage participants to contribute to federated learning.

Finally, this article [16] applies the combination of Blockchain and Federated Learning to smart cities. Their framework, BFLIoT, relies on edge computing to reduce latency and optimize resource utilization while providing advanced protection against cyber attacks.

Existing work on TinyML and federated learning shows that these technologies allow smart devices to process data locally. However, most approaches, like Federated TinyML, still rely on a central server for model aggregation, which creates a vulnerability point. Other studies, in particular, explore integrating blockchain to enhance security and privacy in federated learning [17].

Our approach, TinyFLChain, stands out by proposing a fully decentralized system that combines blockchain and federated learning, designed for TinyML devices through optimized model exchanges and reduced blockchain computation

overhead. Our solution uses smart contracts with specific conditions to reward clients with the best performance and penalizes the worst clients, along with server choices each round to provide an incentive mechanism and penalize clients who do not contribute effectively. In the first round, choices are random, but after that, the selection is based on client evaluations at the end of each round.

IV. PROPOSED FRAMEWORK

In this section, we describe the proposed architecture of TinyFLChain, which details its key components and the incentive mechanism.

Figure 1 illustrates the architecture of TinyFLChain that takes a decentralized approach to federated learning in combination with blockchain and MQTT for communication between TinyML clients. Before participating in the federated learning process, clients have to register on the blockchain through the smart contract. Then, the federated server is selected randomly, but after each learning round a new server is chosen based on client performance and each participant's Ethereum balance. This ensures that all participants are verified and their contributions are securely recorded.

The smart contract is a core component of TinyFLChain, automating key processes and ensuring transparency within the blockchain network. It also plays a vital role in managing the incentive mechanism, ensuring that clients who actively contribute to the learning process are fairly rewarded.

The incentive algorithm 1 rewards the most accurate client at the end of each training round with a proportional amount of Ether, encouraging high-quality contributions. Clients with poor updates receive reduced rewards, promoting active participation and discouraging low-quality training. Although not intended for detecting malicious behavior, the system incentivizes model improvement. Additionally, the aggregation server receives a fixed Ether reward to ensure fairness across roles.

Additionally, the global model weights are stored on the blockchain to ensure recoverability in case of data loss. MQTT facilitates efficient and secure communication among clients by enabling the exchange of global and local model weights through predefined topics, supporting fast and reliable model update transfers.

V. EXPERIMENTAL SET UP

To evaluate our framework, we deployed it on Raspberry Pi 4 Model B devices, each acting as a node to simulate the integration of federated learning with blockchain in a decentralized environment. Each device featured 4GB of RAM and a 16GB Class 10 microSD card, running Raspberry Pi OS on a 64-bit Linux architecture. This setup allowed us to assess the practicality of executing both federated learning and blockchain operations on resource-constrained edge devices.

The software architecture of our system, illustrated in Figure 2, integrates federated learning and blockchain technologies. Ganache serves as the local Ethereum blockchain

environment, while smart contracts are developed and deployed using Solidity. MetaMask ensures seamless interaction with the blockchain. Federated learning is implemented using Python and TensorFlow [18], enabling efficient training and aggregation of local models. Communication between system components is handled through the lightweight MQTT protocol.

Algorithm 1 Incentive Strategy for Enhancing Federated Learning Contribution

Input: List of clients $\mathcal{C} = \{(a_i, \text{acc}_i)\}$, total reward $\mathcal{R}_{\text{total}}$, penalty $\mathcal{P}$, bank account B
Output: Distribution of rewards and penalties
Sort clients by accuracy in descending order
Extract accuracies: $A = [\text{acc}_i]$
 $A_{\text{total}} \leftarrow \sum A$
if $A_{\text{total}} = 0$ **then**
 return no reward given
end if
Compute contributions: $c_i \leftarrow \text{acc}_i / A_{\text{total}}$
for each client i **do**
 $r_i \leftarrow \lfloor \mathcal{R}_{\text{total}} \cdot c_i \rfloor$
 if $r_i > 0$ **then**
 Transfer r_i ETH from server to client i
 end if
end for
Identify worst client $w \leftarrow \arg \min \text{acc}_i$
if $\text{balance}(w) \geq \mathcal{P}$ **then**
 Transfer $\mathcal{P}$ ETH from client w to server
end if
/ Reward the server for aggregation */*
 $\mathcal{R}_{\text{server}} \leftarrow 0.05 \text{ ETH}$
if $\text{balance}(B) \geq \mathcal{R}_{\text{server}}$ **then**
 Transfer $\mathcal{R}_{\text{server}}$ ETH from bank B to server
end if

We evaluated TinyFLChain's performance using three widely benchmarked image classification datasets: CIFAR-10, MNIST, and Digits. These datasets were selected to cover a range of classification challenges, including image size, complexity, and scale variations, allowing us to assess the system's adaptability to simple and complex tasks.

- **CIFAR-10 dataset:** [19] contains 60,000 color images of 32×32 pixels, categorized into ten classes. Each class has 6,000 images, with 50,000 of them for training and 10,000 for testing. This dataset is chosen because of its complexity, which includes real-world objects.
- **MNIST dataset:** [20] which contains 70,000 grayscale images of handwritten digits ranging from 0 to 9, is another dataset used in image classification research. The size of each image is 28×28 pixels, with 60,000 for training and 10,000 for testing.
- **Digits dataset:** [21] from the scikit-learn library. The dataset has 1,797 grayscale images of handwritten digits, each with a resolution of 8×8 pixels.

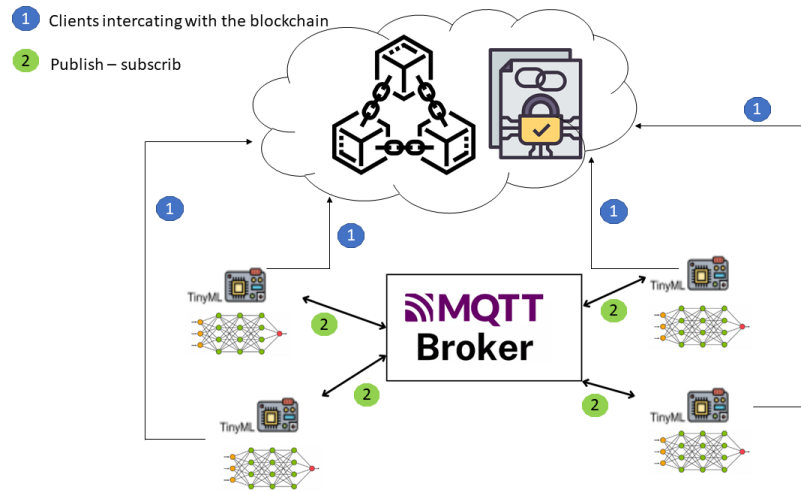


Fig. 1. TinyFLChain:Decentralized Federated Learning Architecture with Blockchain and MQTT.



Fig. 2. TinyFLChain: Software architecture components.

VI. RESULTS ANALYSIS AND DISCUSSION

Our system's evaluation is presented through figures illustrating how accuracy evolves over 10 training rounds. All figures have an X-axis representing the number of rounds, while the Y-axis displays the model's accuracy across different datasets.

Before evaluating our proposed system, we developed a centralized and a decentralized federated learning architecture. We used the MQTT protocol in both approaches to efficiently exchange global and local weights. In order to evaluate performance, we tested 10 training rounds using accuracy as our primary metric.

The results obtained for the centralized federated architecture are presented in Figure 4. The evaluation was performed on three selected datasets. During the 10 training rounds, there was a gradual improvement in accuracy, highlighting the effectiveness of the centralized federated learning. By the ninth round, the accuracy of the MNIST dataset reaches 100%, demonstrating rapid convergence and excellent gener-

alization on this relatively simple dataset. For CIFAR-10, a more complex dataset, the final accuracy reaches 87%, which remains a significant advantage in a federated context. For the Digits dataset, the accuracy reaches 97% after 10 rounds, proving the model's robustness in classifying handwritten numbers.

Figure 5 shows how accuracy evolved over 10 rounds for three datasets in a decentralized federated learning context. MNIST shows a rapid increase in accuracy, reaching 99.25% in the 10th round. However, CIFAR-10's improvement is more gradual, from 35.15% in the first round to 38.90% in the last round, suggesting that federated training has more difficulty generalizing complex data. Finally, The accuracy of the Digits dataset is consistently 10.72% throughout the rounds, which suggests a lack of significant learning.

The results of TinyFLChain architecture shown in Figure 6 show the evolution of the accuracy of the best and worst clients for MNIST and CIFAR-10 datasets in a decentralized architecture based on federated learning with blockchain. As previously discussed, our system implements an incentive mechanism. Figure6, reveals the impact by comparing the performance of the best and worst clients. The accuracy of the best client for MNIST is generally stable at 85.75% to 87.18%, while the worst client's accuracy varies between 83.24% and 84.93%. This indicates that even in a decentralized environment, performance remains consistent across customers. CIFAR-10 significantly improves accuracy for the best client, from 50.70% to 62.09%, while the worst client's evolution is slower, from 49.12% to 53.01%. The higher complexity of CIFAR-10 images may be the reason for this disparity, which makes learning more difficult and increases the impact of differences between clients.

In this study, we used the same model for all three selected datasets. Despite this, the results obtained differ due to the features of each dataset, which highlights the

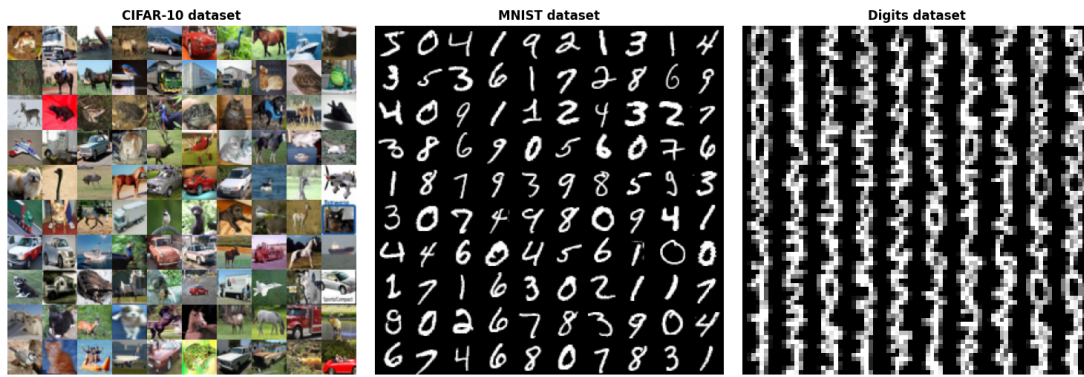


Fig. 3. Sample images from CIFAR-10, MNIST, and Digits datasets.

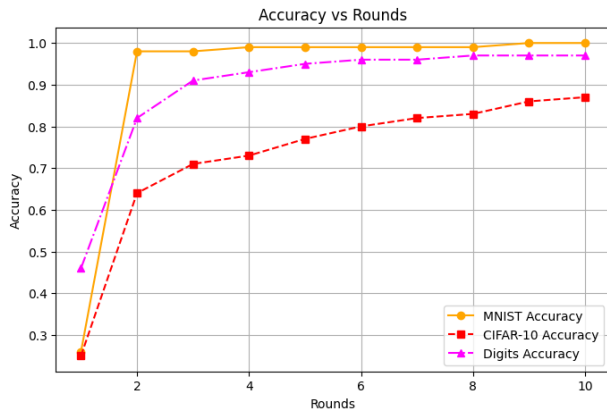


Fig. 4. Accuracy progression over 10 rounds in Centralized Federated Learning (MNIST, CIFAR-10, Digits).

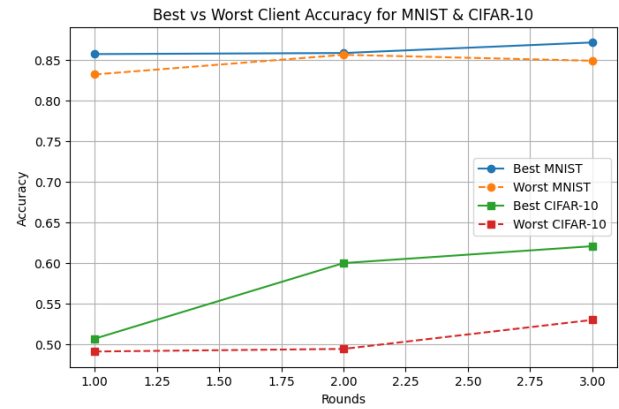


Fig. 6. Accuracy of best and worst clients using TinyFLChain on MNIST and CIFAR-10

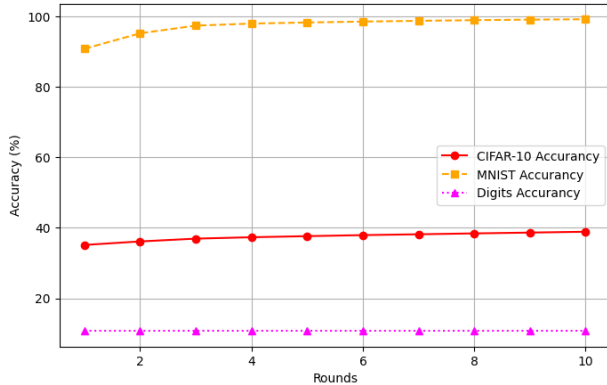


Fig. 5. Accuracy evolution across 10 rounds in decentralized FL (MNIST, CIFAR-10, Digits).

importance of choosing a model adapted to the complexity and specificities of each dataset. For the systems we tested, we clearly observed that we achieved better results with the MNIST dataset. The reason for this is due to the specific characteristics of MNIST, simple and well-defined shapes, low intra-class variability. MNIST is easier to classify than CIFAR-10, which has more complex. On the other side, the

poor performance obtained on the Digits dataset indicates that the architecture used is not suitable for small-sized images (8×8). Indeed, a more compact convolutional network with fewer layers and reduced filters, or a lightweight multilayer perceptron, would be more appropriate to this type of data. The use of data augmentation techniques could also enrich variability and improve accuracy. Finally, the exploration of more suitable federated aggregation strategies would help to improve generalization on constrained datasets.

The integration of blockchain is one of the major strengths of our architecture, which ensures greater transparency and security. In reality, the model weights are retained in the blockchain, ensuring their integrity and preventing any unauthorized modification. It is essential to consider resource management in the decentralized federation architecture with blockchain. Unlike a centralized approach, which allows for the optimization of resources in a global way, the distributed architecture must adapt to the individual capabilities of each client, which can lead to constraints. During our experiment, after three training rounds, the system slowed down due to overloading of the Raspberry Pi devices, which caused RAM to saturate and led to the process stopping. The issue was made worse because one Raspberry Pi node was running both

blockchain node and Federated Learning computations, significantly increasing resource consumption. The blockchain layer's computational overhead caused a decrease in model accuracy because of slower training and communication delays. Although there is a performance trade-off, integrating blockchain can provide enhanced security and trust, ensuring data integrity and resisting malicious modifications in the federated learning process.

VII. CONCLUSION

This paper introduces TinyFLChain, a secure and optimized Federated Learning architecture for IoT devices, leveraging TinyML and blockchain. It uses MQTT for lightweight communication, ensuring data security and decentralization, while smart contracts manage model parameters transparently. Experimental results show good performance across varied datasets, although resource management issues, like system slowdown on Raspberry Pi devices after three training rounds, highlight the need for optimized resource allocation. Future work will also focus on optimizing blockchain efficiency through lightweight consensus mechanisms and off-chain storage and measuring energy consumption, before testing TinyFLChain under adversarial conditions to assess its robustness and identify potential weaknesses in the incentive mechanism.

VIII. ACKNOWLEDGMENT

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